Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Digital Twin for Tokamak and Stellarator Design and Optimization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources			
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	*Next-Generation Nuclear Power*		
Concept of Proposed Technology	**The project aims to build digital twins for tokamak and stellarator design and optimization.** This proposed research is built upon years of support of the PI's development in fusion whole device modeling and AI/ML techniques. This includes our 6-year Tokamak Disruption Simulation DOE SciDAC center ($18M, https://tds-scidac.github.io/) and a 5-year DeepFusion AI/ML initiate ($4M). Both were supported by DOE Fusion Energy Sciences Program. It also leverages two ASCR ML/AI projects for which Dr. Tang is the lead PI and a large ASCR MMICC center (https://charmnet-mmicc.github.io/).			

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Rationale for Joint R&D Project	A private fusion startup who has strong interest in design and optimization for tokamak, stellarator or other magnetic confinement fusion devices shall be well correlated with this project. This project will be significantly benefited from a close collaboration with a private company by adapting the developed techniques for practical design need and experimental data.
Key Words	1. MHD equilibrium 2. Tokamak and stellarator 3. Design and optimization 4. Digital twin 5. Data assimilation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]		**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives		The project has the objectives of (1) delivering advanced scalable high-performance computing solvers for Tokamak and stellarator equilibrium; (2) developing advanced machine learning (ML) techniques as digital twins for magnetic confusion devices' control and optimization; (3) develop a data assimilation technique to incorporate experimental data or first-principles simulation data to further improve the fidelity of the proposed digital twins.			
R&D Target Specifications		**Task 1:** We will enhance our existing tokamak Grad–Shafranov solver [1, 2] in three ways: (i) develop advanced meshing and adaptivity technique to better suit the existing tokamak design and optimization; (ii) extend the solver to include equilibrium reconstruction capabilities; (iii) extend the solver to a 3D equilibrium solver for stellarator **Task 2:** We will develop surrogates for reconstructing plasma states in tokamaks and stellarator. We aim to (i) develop structure-preserving surrogates for solution maps associated with underlying system and (ii) train using synthetic data from our forward solvers or experimental data. **Task 3:** We aim to develop a probabilistic, structure-preserving neural network for real-time equilibrium reconstruction from diagnostics. The objective is to produce physically admissible equilibria (not merely sensor-consistent states) while maintaining Bayesian/ensemble uncertainty on quantities. The deliverable is a real-time inference capability supported by solver-trained surrogates for streaming DT operation.			
R&D Plan		Tasks and milestones: Taks 1: • Y1: Develop an adaptive, 2D equilibrium solver that supports separatrix tracking and scalable solvers and adapt our solvers for the company's need of tokamak configuration or other MCF devices. • Y2: Extend the existing solver for equilibrium reconstruction that supports various diagnose in the optimization loop. • Y3: Extend the existing solver to be integrated with other physics models such as sources due to runaway electrons for disruption prediction and mitigation. Task 2: • Y1: Define the structure-preserving machine learning surrogate for equilibrium reconstruction and demonstrate structure preservation with baseline accuracy benchmarks. Examples of our structure-preserving ML plasma applications can be found in [3, 4]. • Y2: Develop a physics-informed super-resolution operator, form an end-to-end surrogate, and evaluate accuracy, speedup, and robustness for forward solving and reconstruction. • Y3: Stress-test under out-of-distribution conditions, deploy in forward solve surrogate and reconstruction workflows, and deliver validated, reproducible software with documented performance.			

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		Taks 3: • Y1: Deliver a baseline generative model for the joint p(x,y), and demonstrate conditional sampling for p(x | y) with partial diagnostics, including baseline calibration and out-of-distribution diagnostics. • Y2: Integrate the generative model into the digital twin loop with real-time conditional sampling; Validate coverage under diagnostic dropout and domain shift in streaming-style tests. • Y3: Verification and validation, particularly for the need of company and their available datasets from experiment or numerical simulations.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research group of Dr. Tang at Georgia Tech consists of 4 PhD students, 2 master's students, and several undergraduate students. He is also in several projects with the lead experts in fusion modeling from Los Alamos, Livermore, and Oak Ridge National Labs. We will be responsible for developing solvers, ML techniques, modifying the algorithm design based on the fusion startup's need, incorporate the experimental (such as KSTAR) or simulation data provided from the company, and verification and validation (VV) of our algorithms.
Expected R&R of Korean SME Partner	The Korean SME shall provide the need of their specific design (tokamak, stellarator, or other magnetic confine fusion devices where equilibrium reconstruction is needed), work closely with our team to provide the guidance in fusion physics, and provide data including examples such as KSTAR from Korea Institute of Fusion Energy or their forward high-fidelity simulations.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Fusion energy provides a promising solution as a next-generation nuclear energy. At US and China, it has attracted significant amount of public and private investment in the past few years. The proposed research is well aligned in the design and optimization of magnetic confinement fusion devices. A reliable equilibrium reconstruction is central to plasma control, disruption avoidance, scenario development, and ultimately steady high-performance operation. Moreover, there is a rapidly growing emphasis on AI for Fusion: the field is moving toward data-rich, high-rate sensing and learning-augmented control, but progress is bottlenecked by trustworthiness. By anchoring inference and forecasting to physical structure, this project aims to deliver a digital twin workflow that can be tuned for private fusion startup's needs and accelerate the arrival of commercial fusion.

6. References

(1) Serino, D. A., Tang, Q., Tang, X. Z., Kolev, T. V., & Lipnikov, K. (2025). An adaptive newton-based free-boundary Grad–Shafranov solver. SIAM Journal on Scientific Computing, S364-S385.

(2) Liu, S., Tang, Q., & Tang, X. Z. (2021). A Parallel Cut-Cell Algorithm for the Free-Boundary Grad--Shafranov Problem. *SIAM Journal on Scientific Computing*, *43*(6), B1198-B1225.

(3) Drimalas, E. G., Fraschetti, F., Huang, C., & Tang, Q. (2025). Symplectic neural network and its application to charged particle dynamics in electromagnetic fields. *Physics of Plasmas*, *32*(10).

(4) Burby, J. W., Tang, Q., & Maulik, R. (2020). Fast neural Poincaré maps for toroidal magnetic fields. Plasma Physics and Controlled Fusion, 63(2), 024001.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization